

Supplementary Information (integration draft)

Status: portfolio-level disposition is imported from NeuroMech Route-D evidence matrix V3 (39 records). This supplement includes the complete 39-row portfolio disposition and V1 run records, but it is not yet a fully adjudicated source-to-claim supplement. Selected records from the divergent experiment-publication branch are listed separately in ../evidence/PUBLICATION_BRANCH_CROSSWALK.csv. C04–C06 have selected-file hash pins but no exact V3 row. C08's pinned contract identifies it as a post-result follow-up informed by, and distinct from, M5_ELIGIBILITY_TEMPORAL_XOR_V2; its relationship to the M10 record remains unresolved. Individual selected-file hashes and a passing stored-output verification are reported, while the package-root checksum list retains stale/unavailable entries. The C08 evidence identity is therefore narrowed but the full source-package audit is not closed.

Supplementary Table S1 — Full Route-D portfolio disposition

../supplementary/TABLE_S1_PORTFOLIO_DISPOSITION.csv contains all 39 rows from PROJECT_EVIDENCE_CLASSIFICATION_MATRIX_V3.csv, with an added proposed NeuroConverge disposition. It preserves the original labels, portfolio scope, units, source pointers and interpretation limits. The imported source matrix hash and source worktree state are recorded in ../workstreams/PAPER_CRITICAL_SOURCE_SNAPSHOT.json.

The portfolio includes the bounded M0 assay; Fish1.5 primary, invalid-null, robustness and post-result records; published worm evidence and schema-limited analyses; M2 synthetic rounds and historical failure; M4–M11 algorithmic studies; M12 task-viability failure; Drosophila-inspired synthetic tests; and separately governed RR18/RR19 records. Different studies are not replications and are not pooled. RR18/RR19 remain separate workstreams.

Supplementary Table S2 — Publication-branch crosswalk

../evidence/PUBLICATION_BRANCH_CROSSWALK.csv maps the nine claims from the initial NeuroConverge draft to exact or unresolved Route-D matches. The disposition is: C01–C03 map to V3 main-text candidates; C04–C06 are identified branch packages retained in the Supplementary Information without exact V3 rows; C07 is an editorial synthesis, not a source record; C08 is a post-result follow-up whose relationship to M10 remains unresolved; and C09 maps to a V3 supplementary-only record. “No exact row” means there is no exact counterpart in the 39-row V3 matrix. It does not imply that selected files are unidentified. No branch-only package is merged into V3 disposition counts. RR18/RR19 are present as separately governed entries in the V3 disposition inventory but excluded from this synthesis; their appearance in that inventory is not an inclusion as evidence for the manuscript.

Initial-draft claim group	Count	Manuscript disposition	Selection rationale / unresolved boundary
C01–C03	3	Main-text evidence candidates	Exact V3 matches for the bounded connectome assay, Fish1.5 records and published worm evidence/schema audit; distinct evidence types remain separated.
C04–C06	3	Supplementary branch records	Selected packages are identified and hash-pinned, but have no exact row in V3; retained for transparency without merging into the 39-row counts.
C07	1	Organizing synthesis in main text	A framing proposal, not an empirical record or validated hierarchy; requires novelty assessment.
C08	1	Supplementary post-result follow-up; M10 relationship unresolved	Its pinned contract describes a distinct 32-seed/five-arm follow-up informed by the earlier 30-seed/three-arm M5_ELIGIBILITY_TEMPORAL_XOR_V2 (C08 contract ; V2 summary). The pinned publication branch does not provide the M10 final-status artifact needed to resolve that relationship. Selected-file hashes and stored-output verification do not cure stale/unavailable entries in the C08 root checksum list.
C09	1	Supplementary V3 record	Exact V3 record exists but is classified supplementary-only in the source matrix.
RR18/RR19	—	Separately governed exclusions	They are recorded in the inventory where applicable, but excluded from this manuscript synthesis; no ownership boundary is changed here.

This accounting closes the disposition of the nine initial-draft claim groups for reporting purposes. It also resolves one earlier ambiguity: C08 is a separate post-result follow-up to V2, not the V2 result itself. The C08-to-M10 relationship and full package checksum audit remain open; C08 must remain supplementary and must not support an M10-linked claim until those source records are available and reconciled.

Supplementary Methods S1 — Retrospective portfolio selection

This integration is a retrospective, source-linked synthesis, not a systematic review or prospectively unified biological experiment. The main-text cases were selected to illustrate distinct evidence transitions and include both supported and unsupported outcomes; this selection was made with knowledge of earlier project outcomes and is not an outcome-independent sample of the portfolios. The initial synthesis and M2/M5 tests were followed by a post-result DMP fast-slow $\times$ correspondence module and a source-informed V1 PC-SST figure/ground package. Each local protocol was frozen before its own outcome run, but all were designed after earlier project results had been inspected; none is project-level preregistration or independent biological confirmation. The V1 protocol's AP interaction interpretation had a sign-direction wording error, preserved and disclosed in its post-run amendment. Each Route-D portfolio record retains its original evidence class, unit, task, comparator, status (including invalid, blocked, or outcome-informed), and correction history in Table S1. Selected branch records are accounted for separately in Table S2; unresolved C08-to-M10 identity is not used to support a claim. A record's presence in the portfolio matrix does not imply that it supports the central claim. Supplementary-only and excluded records remain visible in the disposition accounting, including RR18/RR19 as separately governed exclusions. No common effect or meta-analysis is calculated because the tasks, outcomes, units and sampling structures are not exchangeable.

Supplementary Methods S2 — Source provenance and reproducibility boundary

The NeuroMotif public main candidate is 1adc07ffc735cf458116ca36c9f4875708034d4a. The NeuroMech public experiment-publication candidate is 3e5c458f15b2cad7aa6c5703b1827763e84343bb; its local worktree had two changed/untracked paths at audit. The broader local Route-D view is based on 555c064f4ff3153c871808c3f191b9934a18eaf5 plus 387 changed/untracked paths. These states are not interchangeable and none is an owner-approved immutable full freeze. Paper-critical file hashes are in `../workstreams/PAPER_CRITICAL_SOURCE_SNAPSHOT.json`; this proves file identity at audit time, not that a result was independently rerun or redistributable.

Source files and historical environments remain heterogeneous. Exact runners, dependency locks, licenses and redistribution rights must be evaluated per selected record. The integration repository is not a replacement for either source repository and does not claim full independent reproducibility.

Supplementary Methods S3 — Cerebellar source-readout reanalysis

This post-result exploratory analysis used the released Dryad v4 file `learning_1s_to_2s_GrC_CF.mat` [10] and source-code-derived plasticity computation. All sessions in the three released groups were included: five 1-s expert sessions, five 2-s novice/1-s expert sessions and six 2-s expert sessions. Eligible rewarded trials satisfied the source `mvlen > 7` and `goodmvd` criteria. Each session used five deterministic random 50/50 trial splits, with seed `20261001 + 100000g + 1000s + k` using zero-based group, session and split indices. Climbing-fibre selection, activity scaling and readout weights were fitted only on training trials.

Candidate climbing fibres showed a reward-aligned event increase in $[0, 0.25]$ s relative to $[-0.3, -0.025]$ s. The source plasticity computation used granule-cell activity in $[-0.15, -0.025]$ s before teaching events, with rounded frame offsets and inclusive window endpoints. Activity was rectified, scaled by training-cell 95th percentiles, transformed with the source logistic function, averaged and normalized per selected climbing fibre, then averaged with the source negative sign. These weights were modeled proxies under the inherited connectivity assumption. Held-out readouts used movement-midpoint-aligned activity through the recorded reward delay, capped at 2 s, and squared Pearson correlation with elapsed time. Uniform, cell-shuffled and timing-shuffled controls and a training-standardized ridge reference with $\alpha = 1.0$ were evaluated on the same held-out trials. Valid trial scores were averaged within split, splits within session and sessions within source group. Two splits without training climbing-fibre candidates remained unusable. No animal-level inference was calculated, and the ridge reference was not capacity matched.

The exact package is pinned to NeuroMech experiment-publication commit 3e5c458f15b2cad7aa6c5703b1827763e84343bb and selected-file hashes in `../reproducibility/SELECTED_ARTIFACTS.csv`. Route-D V3 has no exact record for this package and classifies the broader M5_CEREBELLAR line as supplementary-only; this analysis is therefore reported as a separate supplementary branch record and is not included in the 39-row Route-D disposition counts.

Supplementary Methods S4 — DMP fast-slow correspondence V1

The DMP V1 synthetic extension was motivated by the published fast-slow dual-memory spiking architecture of Sun et al. [13], but was not a reproduction of its algorithm or hardware. Its artificial model combined 32 leaky integrate-and-fire units (leak 0.5, threshold 1, subtractive reset and fixed fast-sigmoid surrogate gradient) with a fixed two-dimensional LMU-like state. The Legendre-derived continuous matrices were discretized by exact zero-order hold at horizon $\theta = 32$. The short-state control retained the same state dimension, fixed input matrix, learnable tensor shapes, initialization scheme, optimizer and update count, but used a 0.5I transition. Slow-state output entered the membrane current through a learned scalar-compressed input. DMP and control parameter counts were 1,286/1,385 for regression/classification; an 18-unit GRU reference had 1,315/1,372 parameters, within 5% of DMP. FLOPs were not matched.

Each generator used four input channels and a 64-step training sequence. In delayed association, a noisy four-level context occupied steps 0–3 and a noisy four-level query occupied steps 36–39; a fixed task-specific permutation of (context + query) mod 4 supplied the class. For state estimation, a stationary unit-variance AR(1) latent trajectory (φ sampled from 0.90–0.98 per task instance) was observed through two channels; a binary context indicated which channel had low (SD 0.25) or high (SD 1.2) noise, and target was the final latent state. Context regime switch probability was sampled from 0.015–0.045. In contextual decision, a four-level context at steps 0–3 selected a task-specific mapping from four noisy evidence votes at steps 36–51 to a class. Longer-delay test sequences had 128 steps: query or evidence moved to steps 68–71 or 68–83, respectively; state-estimation horizon doubled. Configuration seeds were shared across IID and longer-delay generator instances, while trial draws differed.

Each task-instance seed generated 2,048 training, 512 IID test and 512 longer-delay test trials. Training prefixes $N = 128, 512$ and 2,048 were nested. For every prefix separately, the slow-context stream was reassigned by a no-fixed-point derangement within that prefix; IID and longer-delay splits used independent derangements. This preserved the exact context-stream marginal for each comparison while holding the fast stream, targets and sample count fixed. The intervention broke trial-level pairing and did not change target labels. Two paired initializations were averaged within each task-instance seed.

All learned models used fixed ten epochs, batch size 64, Adam learning rate 0.003, gradient clipping at five, and the same task data and update budget. The three families used 30 task-instance seeds each. Classification endpoint was test 0–1 loss; state endpoint was test squared error against the unit-variance latent state. The co-primary contrast was $I = [L(M+, C-) - L(M+, C+)] - [L(M-, C-) - L(M-, C+)]$, where $M+$ is DMP, $M-$ is the short-state control, $C+$ is aligned and $C-$ is broken correspondence. Each family had a paired task-seed percentile bootstrap 95% interval and two-sided sign-flip P value; Holm correction controlled the three co-primary interactions. Task viability required aligned, full-data GRU loss to be below task-null loss; the three viability P values were also Holm-adjusted. A positive mechanism-specificity gate required a passed viability gate, positive adjusted interaction evidence and a positive aligned DMP advantage. Secondary sample-efficiency analysis estimated the within-unit slope of aligned control-minus-DMP loss against $\log_2(N)$; its three family P values were Holm-adjusted. The gate required the $N=128$ advantage interval above zero, slope interval below zero and adjusted $P < 0.05$. Longer-delay changes were secondary and evaluated without refitting.

The runner, verifier and contract hashes and runtime environment were frozen before confirmatory units. Runtime was Python 3.9.6, NumPy 2.0.2 and PyTorch 2.8.0 on macOS arm64 CPU, deterministic algorithms enabled and one thread. The run produced 7,020 unit-level rows and 7,020 per-trial loss vectors. The independent verifier recalculated all means from raw vectors and checked hashes, task-instance coverage, parameter counts, cell completeness and exact correspondence marginals. The module is a post-result project extension because prior M2/M5 outcomes and the fast-slow literature had been reviewed before it was designed.

Supplementary Methods S5 — Prospective source-informed V1 figure/ground test

The source paper reports like-to-like functional connectivity from orientation-tuned pyramidal cells to SST interneurons and a causal contribution of SST activity to orientation-dependent surround suppression [14]. The present benchmark did not use recordings from that study. Each 11×11 synthetic image contained eight orientation-tuned response channels. A fixed soft tuning function ($\kappa=2$) generated channel responses with Gaussian noise (s.d. 0.08), positive clipping and normalization. One randomly placed 3×3 figure had an orientation category selected uniformly from the seven alternatives to the background category. Training and IID backgrounds were uniform; the OOD split used the fixed category probabilities [0.35, 0.25, 0.12, 0.08, 0.07, 0.05, 0.04, 0.04]. Each of 30 deterministically derived task-instance seeds generated 512 training images, 256 IID test images and 256 OOD images. Nested training prefixes were 32, 128 and 512 images.

At each location, surround activity was the mean of the eight neighbouring locations. The feature-selective transform subtracted 0.65 times the same-channel surround response. In the broken-correspondence arm, channels were routed through a fixed cyclic no-fixed-point permutation. The global-pool control subtracted 0.65 times the across-channel

mean surround response, broadcast to all channels; it is mathematically invariant to the permutation. A regularized linear discriminant (ridge $\lambda=0.02$) was fit separately in each factorial cell using the same training images. A generic-context reference received concatenated center and surround channels (16 features) with the same readout; it was not parameter-matched and was not a strong neural-network reference. The null AP was the positive-pixel prevalence, 9/121.

The independent unit was the task-instance seed; image and pixel measurements were nested. The primary endpoint was mean per-image average precision. The frozen interaction was $(AP(\text{selective, broken}) - AP(\text{selective, aligned})) - (AP(\text{global, broken}) - AP(\text{global, aligned}))$. The contract text incorrectly described a positive value as an alignment benefit even though AP is higher-is-better and the contrast is broken-minus-aligned; the numerical formula, outputs and gate were left unchanged. The positive-specificity gate required generic-reference viability, the frozen numerical interaction criterion and higher aligned AP for selective than global pooling. Paired bootstrap CIs used 20,000 task-seed resamples; two-sided sign-flip tests used 100,000 Monte Carlo sign patterns generated by NumPy default_rng(20261003). Scaling and OOD comparisons were secondary. The executed contract, runner, verifier, derived seed schedule, hashes, 230,400 per-image outcomes and post-run amendment are in `../new_experiments/PROSPECTIVE_V1/`.

Supplementary Results S1 — Full portfolio interpretation

Consult Table S1 for the complete row-level status and ceilings. In brief, the route contains positive, negative, invalid, inconclusive, schema-blocked and supplementary-only records. M2 comprises multiple synthetic/controller rounds and cannot be summarized by the single favorable sensory-site versus output-persistence contrast alone. M5 and related eligibility studies are heterogeneous; V3 classifies the M5 line as supplementary-only, while some selected experiment-publication packages have no exact V3 row. M11 and M12 must be retained in the full portfolio account, as must the *Drosophila* synthetic counterevidence and robustness records. These records constrain project-level selection but do not become evidence for a biological mechanism merely by being included.

Supplementary Table S3 — M2/M5 outcome-informed synthetic V1 tests

M2 and M5 are separate synthetic studies with different outcomes and inferential units. M2 used 32 paired training-seed blocks and mean squared tracking error (MSE); its primary contrast is action-only minus realized-state MSE under hidden reversals, so a negative value favours action-only. M5 used 30 task seeds and normalized mean absolute error (nMAE); its contrast is broken minus aligned correspondence, so a positive value means error increased when input-target correspondence was broken. Intervals resample the corresponding paired seed units. No pooled estimate is defined.

Study and arm	Independent unit (n)	MSE action-only / realized-state or nMAE aligned / broken	Contrast and 95% interval	Interpretation
M2 primary: action-only vs realized-state, hidden reversals	Training-seed block (32)	MSE 2.43809 / 2.50618	Action-only – realized-state = -0.068094 [-0.141077, -0.003541]	The exact execution-state channel did not improve this controller; the contrast favoured action-only for this plant and training budget. It does not identify an information-specific penalty or test the biological RIM→AIY signal.
M5 CF-timed local rule	Task seed (30)	nMAE 0.141521 / 0.258109	Broken – aligned = +0.116588 [+0.115072, +0.118065]	Error increased when correspondence was broken; the contrast was positive in all 30 seeds. This supports use of correspondence by the tested synthetic rule, not a biological mechanism.
M5 full-input replay	Task seed (30)	nMAE 0.071385 / 0.260936	Broken – aligned = +0.189552 [+0.186912, +0.192229]	The correspondence effect was larger than the local-rule effect; replay receives all input bins.
M5 generic ridge on the same 16 features	Task seed (30)	nMAE 0.056643 / 0.259033	Broken – aligned = +0.202390 [+0.200123, +0.204675]	The generic feature-based effect was larger than the local-rule effect. The scalar ridge head, objective and fitting procedure differ from the local rule, so this is not a capacity-matched learning-rule comparison.
M5 no-trace control	Task seed (30)	nMAE 0.258153 / 0.258083	Broken – aligned = -0.000070 [-0.000411, +0.000219]	The point estimate slightly decreased after correspondence was broken; its interval crossed zero and no equivalence margin was specified.
M5 CF-time-shuffled control	Task seed (30)	nMAE 0.259511 / 0.258312	Broken – aligned = -0.001199 [-0.002510, +0.000072]	The point estimate slightly decreased after teaching-time shuffling; its interval crossed zero and no equivalence margin was specified.

The full seed-level outputs, frozen contracts, source/code provenance and independent verification records are in `../new_experiments/`. The M2 and M5 results are not pooled and do not validate either motivating biological mechanism.

Supplementary Table S4 — DMP fast-slow × correspondence V1

The new module was a post-result project extension with a protocol frozen before its 90 task instances were generated. Its fast-slow abstraction was inspired by, but is not a reproduction of, a previously published dual-memory spiking architecture [13]. Each family had 30 independent task-instance seeds; the 2×2 mechanism-by-correspondence interaction was the co-primary outcome. Positive aligned advantage means lower DMP loss than the matched short-state control. Task gates test whether the full-data aligned generic GRU beats the task-null predictor. Values below are recalculated from per-trial arrays by the independent verifier.

Task family	Interaction (95% paired task-seed CI)	Holm-adjusted P	Aligned DMP advantage at N=2,048 (95% CI)	GRU task gate	DMP aligned longer-delay loss change (95% CI)	Mechanism-specificity gate
Delayed association	-0.00212 [-0.00615, 0.00173]	0.64755	+0.00208 [-0.00345, 0.00814]	Fail	+0.00143 [-0.00866, 0.01081]	Not met; viability gate failed
State estimation	-0.00218 [-0.00383, -0.00054]	0.05058	-0.00243 [-0.00363, -0.00123]	Pass	+0.00096 [-0.00187, 0.00375]	Not met; direction favoured control
Contextual decision	+0.00101 [-0.00537, 0.00778]	0.77924	+0.00156 [-0.00573, 0.00931]	Pass	+0.22096 [+0.20153, +0.23841]	Not met; interaction not detected

No family met the positive mechanism-specificity criterion. The state-estimation adjusted P value was 0.05058 and its estimate was negative; it did not meet a 0.05 threshold in the prespecified positive direction. The contextual-decision small-data gate did pass: aligned advantage was +0.07236 at N=128 (95% CI, +0.05036 to +0.09557), +0.03506 at N=512 (+0.01335 to +0.05954), and +0.00156 at N=2,048 (-0.00563 to +0.00941); slope was -0.01770 per $\log_2(N)$ step (95% CI, -0.02288 to -0.01274; Holm-adjusted P = 0.000060). This secondary, single-family result did not satisfy the primary mechanism-specificity gate. Longer-delay differences are descriptive and are not interpreted as broad OOD generalization.

The complete 7,020-row metrics file, 7,020 raw loss vectors, task-instance configurations, pre-run freeze and verification hashes are retained in `../new_experiments/DMP_FAST_SLOW_CORRESPONDENCE_2X2_V1/results/`. The independent verification status is PASS.

Supplementary Table S5 — Prospective V1 PC→SST synthetic figure/ground test

The study was selected after earlier project outcomes and frozen before its own single confirmatory run. AP is higher-is-better; the primary interaction formula below is broken minus aligned, so positive values indicate a relative advantage for broken routing under selective pooling. The contrast's interpretation in the frozen contract narrative had the sign reversed. Intervals resample the 30 task-instance seeds. The full raw per-image AP output and verifier-derived task-seed means are retained in `../new_experiments/PROSPECTIVE_V1/results/`.

Endpoint/contrast	Estimate	Paired 95% CI	Two-sided sign-flip P	Interpretation
Generic context minus prevalence null, N=512 IID	+0.37219 AP	+0.36046 to +0.38379	0.000010	Task viability passed
Selective minus global, aligned, N=512 IID	-0.02208 AP	-0.02737 to -0.01664	0.000010	Aligned selective pooling performed worse
Selectivity × correspondence, N=512 IID	+0.01603 AP	+0.00815 to +0.02357	0.000600	Opposite to the stated alignment-benefit prediction
Selectivity × correspondence, skewed-background OOD	+0.01692 AP	+0.00553 to +0.02913	0.007250	Secondary; same direction as IID
Selective-minus-global aligned slope vs $\log_2(N)$	+0.00091 AP	-0.00108 to +0.00291	0.376247	No reliable sample-size trend

The full positive mechanism-specificity gate failed because the aligned feature-selective arm did not outperform the global-pool control and the interaction direction contradicted the stated prediction. The verifier returned `integrity_pass=true`, with all 230,400 per-image rows, all 180 seed $\times$ sample-size $\times$ split groups, frozen hashes and global-pool permutation invariance verified. The post-run amendment records the sign-direction wording error and the limited linear comparator. The experiment used generated population activity rather than neural observations, and the source paper's Zenodo archive was not downloaded or analysed.

Supplementary Table S6 — Comparator matching and claim limits

Matching is specific to the contrast being interpreted. “Not measured” means that the retained protocol/output does not establish a compute match; similar wall-clock time or parameter count must not be substituted for a FLOPs or inference-compute audit.

Study / comparison	Input information	Trainable capacity	Training / optimization budget	FLOPs or inference compute	Extra task knowledge	Defensible comparison
M2 V1, action-only vs realized-state AcRKN	Same six-channel interface; the auxiliary channel is invalid/zero for action-only and carries the previous executed-displacement sign for realized-state. All remaining observations, flags and commands are shared.	Matched at 320 trainable parameters per AcRKN arm.	Matched: 160 Adam updates, same learning rate, batch size, sequence length, seed pairing and environment streams.	Not profiled or matched.	Neither learned arm receives future/current unexecuted reversal information.	Isolates the added realized-state channel within this AcRKN implementation and budget. It does not establish an intrinsic information penalty or test the biological RIM→AIY signal.
M2 V1, AcRKN vs GRU reference	Both learned families receive the same six inputs.	Near-count matched, 320 vs 323 parameters; architecture differs.	Same update count and training stream protocol.	Not profiled or matched.	No explicit task-distribution oracle in either learned model.	Contextualizes performance; does not isolate an architecture effect.
M2 V1, learned models vs particle filter	The particle filter uses the task's observation/actuation distributions; in the revealed condition it also receives preceding reversal events.	Not capacity matched; particle count is not comparable to trainable parameter count.	Particle-filter gains selected on disjoint validation seeds; learned models use gradient training.	Inference compute not matched; particle counts 64–256 were examined, with 256 used in the main evaluation.	Yes. It uses known task dynamics/distributions.	A task-aware reference point/context only, not a fair architecture or parameter comparison.
M5 V1, CF-timed local rule vs generic RBF ridge	Same 16 temporal RBF features plus bias.	Not matched: local softmax rule has 476 parameters; scalar ridge has 17. The frozen “parameter-matched” label was corrected to “feature-matched.”	Not matched: local rule updates once per trial after complete-trial feature construction; ridge is a closed-form fit with a different scalar regression objective. The local rule is not an online within-trial trace implementation.	Not profiled or matched.	Both use the same supervised targets; neither is given biological observations.	Shows that a generic learner can exploit the same feature representation more effectively in this task. It cannot identify a learning-rule-specific advantage.
M5 V1, CF-timed local rule vs full-input replay	Local rule uses compressed 16-basis full-trial features; replay receives all 40 input bins.	Not matched: 476 vs 41 fitted parameters.	Not matched: one-pass local updates vs closed-form fitting.	Not profiled or matched.	Replay has access to the complete input trial and target labels during fitting.	A stronger-information algorithmic reference; the larger correspondence effect does not isolate mechanism identity.
DMP V1, DMP vs short-state control	Same task inputs, targets and context stream; aligned/broken correspondence is crossed within each architecture.	Matched within the factorial contrast: same learnable parameters, input matrix and state dimension; transition dynamics differ.	Matched within study: paired initializations, ten epochs, batch size 64, same Adam rate and task-seed protocol.	FLOPs explicitly not matched.	No external task oracle is specified for this model contrast.	The primary interaction tests whether the tested slow-state transition changes sensitivity to correspondence, within these task generators. It does not prove biological mechanism identity.

Study / comparison	Input information	Trainable capacity	Training / optimization budget	FLOPs or inference compute	Extra task knowledge	Defensible comparison
DMP V1, DMP vs generic GRU viability reference	Same task families and target definitions; architecture-specific internal states differ.	Near-count matched within 5%.	Same epochs, batch size, learning rate and paired-initialization scheme.	FLOPs not matched.	GRU is a generic task-learning reference; the task gate compares its aligned full-data performance with a task-null baseline.	Viability/context check, not a compute-matched superiority benchmark.
Prospective V1, selective vs global pooling factorial cells	Same synthetic images and orientation channels; both use local/surround context, with selective routing preserving or permuting channel correspondence and global pooling invariant to that permutation.	Matched among four factorial cells by the same eight-input pathway and ridge readout.	Same ridge-discriminant fit and $\lambda = 0.02$ across factorial cells.	Not profiled or matched.	No source neural recordings or external task oracle; all arms use synthetic training labels.	Tests the specified selective-vs-global pooling contrast on this synthetic task only.
Prospective V1, factorial cells vs generic-context reference	Generic reference concatenates local and surround channels (16 inputs); factorial pathway uses eight pooled/routed channels.	Not matched.	Same ridge-regularized linear-discriminant fitting family and λ , but feature dimensionality differs.	Not profiled or matched.	Same synthetic task labels; the reference is linear, not a strong neural-network comparator.	Establishes task viability against a linear context reference, not broad comparator robustness or mechanism-specific utility.

Supplementary Table S7 — Inferential hierarchy across the four synthetic studies

Study	Primary estimand / gate	Independent unit	Multiplicity and inferential status	Claim ceiling
M2 V1	Action-only minus realized-state MSE under hidden reversals; positive would favour the added realized-state input.	Paired training-seed block ($n = 32$); episodes nested.	The primary interval excluded zero in the direction favouring action-only. Paired bootstrap; primary and diagnostic intervals unadjusted, with no portfolio-wide error control or equivalence margin.	A bounded result for this AcRKN, plant and training budget; diagnostics do not isolate an intrinsic information penalty.
M5 V1	CF-timed local-rule nMAE(broken) minus nMAE(aligned); positive means breaking correspondence increases error.	Task seed ($n = 30$); trials nested.	Task viability passed and primary correspondence effect was positive in all 30 seeds. Paired-seed bootstrap; comparator contrasts are descriptive secondary results and are not multiplicity adjusted.	Correspondence mattered to this synthetic rule/task; comparator results do not establish learning-rule specificity.

Study	Primary estimand / gate	Independent unit	Multiplicity and inferential status	Claim ceiling
DMP V1	Three task-family-specific mechanism $\times$ correspondence interactions; separate task-viability gate; secondary small-data scaling gate.	Task-instance seed within family (n = 30); trials/repeats nested.	No family passed the positive mechanism-specificity gate. Holm correction across the three co-primary interactions, separately across three viability tests; scaling slope has its prespecified Holm-adjusted gate. Longer-delay results are secondary/descriptive.	No family passed the mechanism-specificity criterion; the contextual-decision small-data pattern is a narrower secondary result.
Prospective V1	Selectivity $\times$ correspondence AP interaction plus a conjunctive positive specificity gate.	Task-instance seed (n = 30); images/pixels nested.	Task viability passed; the interaction was statistically positive under the frozen formula but opposite to its stated alignment-benefit interpretation, and aligned selective pooling underperformed global pooling. Secondary sample-size/OOD endpoints are not pooled with the primary or corrected across the wider project portfolio.	Synthetic task only. The positive specificity gate failed; do not recode the interaction as an aligned-transfer benefit.

These studies do not share an estimand, unit, endpoint scale or multiplicity family. The intervals and P values must therefore be read within each frozen study contract, not ranked across studies or interpreted as a pooled evidence score.

Supplementary Results S2 — Historical artificial tests and source-readout record

The following numerical account was moved from the earlier main Results or retained from historical source packages. These studies retain their original identities and ceilings; they are not replications of the completed V1 tests. Supplementary Table S2 distinguishes exact hash-pinned branch packages from unresolved cross-branch record relationships. The source-readout endpoint was held-out trial squared Pearson correlation with time progress, a direction-insensitive association rather than calibrated time prediction. Of 80 planned splits, 78 were usable; two retained the status NO_TRAINING_CF_CANDIDATES. Source weights were modeled and sessions were not established as independent animals.

In one synthetic switching task, sensory-site motor feedback reached 88.94% accuracy versus 82.14% for an equal-parameter output-persistence control, but did not exceed no-feedback (89.05%), a two-unit generic recurrent model (90.21%), or a task-aware Bayes filter (90.30%). The local contrast therefore identifies a placement-dependent trade-off against one comparator, not an overall accuracy advantage.

In the cerebellar source analysis, model-derived weights exceeded uniform weights in 15/16 source sessions and CF-time-shuffled weights in 16/16; a fixed ridge readout exceeded the source-derived projection in 14/16. This supports within-session predictive temporal structure under the modeled analysis. In the separate synthetic interval task, CF-timed LTD did not meet its frozen transfer criterion: mean held-out absolute error was 0.1252 s versus 0.1097 s for no-trace, 0.1252 s for CF-time-shuffled teaching, 0.1030 s for a generic radial-basis representation, and 0.0753 s for the training-label empirical timer. A design audit found that trial-specific timing jitter was independently sampled and not present in the inputs. The outcome bounds this implementation/task pair; it does not test whether the biological rule could use a suitable trial-specific signal.

Two additional M5 eligibility-trace studies use different synthetic objectives and must not be combined with the source-defined CF-LTD interval experiment. In a delayed-XOR classification task (32 seeds), eligibility traces reached 0.6974 accuracy versus 0.5562 for no-trace and one-step truncated backpropagation through time (TBPTT-1), but trailed TBPTT-4 (0.9633) and full backpropagation (0.9996). In a separate fixed-generator classification suite, trace eligibility exceeded no-trace by 0.2284 (95% interval [0.2213, 0.2357]); exact replay nevertheless outperformed eligibility in all 12 generator-by-delay cells (overall means 0.7609 versus 0.5590). These are post-result synthetic demonstrations that eligibility can help against a weak memory control under some task conditions; the stronger replay and gradient-based controls set a lower ceiling on claims. Neither result validates a cerebellar mechanism or establishes general transfer.

The M2, M5 source-readout, source-defined transfer, delayed-XOR and task-generator records are pinned to the NeuroMech experiment-publication commit 3e5c458f15b2cad7aa6c5703b1827763e84343bb. Existing stored-output verifiers passed for M2 and the two source-defined CF-LTD packages; stored verification records are available for the delayed-XOR and generator studies, and selected artifact hashes are recorded in `'SELECTED_ARTIFACTS.csv'`. These packages are post-result exploratory at the project level, and the source-readout summaries are descriptive. This pins the cited study records; it does not resolve the broader NeuroMech mainline/worktree freeze or which additional studies belong in the integrated paper.

Supplementary Results S3 — Fish1.5 inference and M0 source boundary

Fish1.5 retained 10,000 valid positive-association permutation draws and 9,844 valid neuron-bootstrap draws out of 10,000 requested draws. The reported bootstrap interval summarizes the valid draws. The frozen topology null preserved binary in/out degrees and the global weight multiset, but not each node's weighted strength. With 861 undefined correlations, its P value remains unestimable. Post-result conditional null analyses do not repair the original frozen inference and retain their separate labels in Table S1. The M0 report documents a crossover only after both retention curves approached noise. Its comparator retained diagonal self-persistence. The complete M0 report is local to the NeuroMech source workspace, and canonical round-artifact linkage is still incomplete; assignment to the structural workstream does not make the full package part of the NeuroMotif public snapshot.

Supplementary Results S4 — M2 V1 diagnostics and reproducibility

The full arm-by-condition means, primary and diagnostic intervals, parameter counts and repeated-run comparison are retained in `../new_experiments/M2_REALIZED_STATE_VS_ACTION_BELIEF_V1/RUN_REPORT.md` and `runs/canonical_1/summary.json`. All 32 seed blocks are included. Both full runs passed the independent verifier and

agree scientifically after only declared timing fields are excluded. The revealed-event control had no frozen equivalence margin; its zero-crossing interval cannot establish equivalent performance. The benchmark received an exact execution-sign measurement, with no tested corruption or delay condition.

The M2 base HEAD 3e5c458f15b2cad7aa6c5703b1827763e84343bb did not define the executed version: the captured contract was modified and the runner/verifier were untracked. SOURCE_PROVENANCE.json identifies the exact contract SHA-256 489ed473bbeb895ccb99ad8b661dbbdeace0f2a19fc6615e1a4dfa8fd6736f0, runner 984ef632108a0aa3256a19009747194818db1296ed0531fe1b294703cf27c8d6, and verifier f6d48727f4ece27f385d56baf1221ac12e09b0cb36ef6e52e6e47d867e7c7055.

Supplementary Results S5 — M5 V1 controls and comparator correction

Full means and effect intervals are retained in

../new_experiments/M5_CORRESPONDENCE_PRESERVED_TIMING_V1/RESULTS.md and results/primary_result.json. The no-trace broken-minus-aligned nMAE effect was -0.000069515 (95% interval $[-0.000411054, +0.000219329]$); the teaching-time shuffle effect was -0.001199416 ($[-0.002510140, +0.000071742]$). Zero-crossing intervals do not establish equivalence or zero information. The same-feature ridge arm has a scalar 17-parameter head, whereas the local rule has a 476-parameter, 28-class softmax head; objectives and fitting procedures differ. The preserved frozen contract and post-run audit record the correction from parameter matched to feature matched. Teaching-time shuffling is a training intervention; the aligned/broken regime independently changes input-target pairing in both train and test splits, so the primary effect does not separate training and evaluation contributions.

The M5 contract SHA-256 is 1434055cc21047edfd8185d50e6dc207c640ddb46aa3cf7b1293a8353c0e7658, and the successful runner SHA-256 is 9f9f4d810f473573fc86f8a8bb2ad454a4d099009004429618d7b766f3bbad80. The first launch failed before producing outcomes because Python 3.9 did not support `zip(strict=True)`; a syntax-only amendment was frozen before the successful run. The original freeze, amendment, execution incident and successful run manifest remain intact.

Supplementary Results S6 — DMP V1 interpretation and verification

Across the three families, no positive mechanism-specificity gate passed. The delayed-association task was marked non-interpretable for the mechanism claim because the aligned generic GRU did not beat the task-null predictor after the prespecified viability analysis. In state estimation, the DMP $\times$ correspondence estimate was negative and its paired interval excluded zero in the negative direction; the Holm-adjusted P value (0.05058) was just above the two-sided 0.05 cutoff. The aligned DMP loss was also higher than the aligned short-state-control loss. This supports a task-bounded disadvantage of this DMP implementation in the tested generator, not a claim that slow state is generally harmful.

In contextual decision, aligned DMP showed a large relative advantage at $N=128$ that diminished at $N=512$ and was not detected at $N=2,048$. Its predeclared slope gate passed, but its full-data mechanism $\times$ correspondence interaction and aligned advantage did not. Moreover, under the longer-delay shift its aligned loss rose by 0.22096 (95% paired task-seed CI, 0.20153 to 0.23841), whereas the delayed-association and state-estimation aligned changes were near zero with intervals spanning zero. This indicates sensitivity to the tested task/shift combination; it does not establish general OOD robustness or a universal sample-efficiency law.

RUN_MANIFEST.json records 30 seeds per family and 7,020 metrics/raw loss vectors. VERIFICATION.json records PASS; file hashes, exact marginals, metric recomputation and complete cells passed. The results remain synthetic and post-result at the project level. Fast-slow memory already has direct prior art, including a 2026 *Nature Machine Intelligence* dual-memory spiking model [13].

Supplementary Results S7 — Prospective V1 interpretation and verification

The source-informed synthetic task passed viability against its pixel-prevalence null, but the feature-selective aligned model had lower AP than the global-pool control. The frozen interaction was statistically different from zero in the direction indicating better performance after correspondence breaking; under the contract's formula and higher-is-better AP, this is opposite to the written alignment-benefit prediction. Therefore, the package does not support positive mechanism specificity or an aligned-routing utility advantage. The same interaction direction appeared under the fixed skewed-background test, while the sample-size slope was inconclusive.

An audit found that the contract's positive interaction interpretation had the sign reversed. Because the issue was identified after outcome generation, no contract, output, model or gate was changed. The formula-defined two-sided result and the failed positive-specificity gate are reported as frozen; the error and limitations are disclosed in ../new_experiments/PROSPECTIVE_V1/POSTRUN_INTERPRETATION_AMENDMENT.md. The selected source/task package

is new to the tested mechanism portfolio but was selected after earlier project outcomes and uses the visual modality already present in adjacent synthetic-visual work. It is not an independent team replication or biological confirmation. The generic context reference is linear and not a strong neural-network alternative.

Remaining assembly work

Study-level procedures for the selected M0, Fish1.5, source-readout, M2 V1, M5 V1, DMP V1 and prospective V1 analyses are now in the main Methods; all six main figures have source-data tables and export hashes. Remaining gaps are the exact source-to-claim crosswalk for some historical branch records, final citation/attribution review, reuse-rights audit and author-supplied end matter. The manuscript remains an internal draft and is not NMI-ready.